Git-hub Link: https://github.com/be25aar/Machine-Learning-and-Neural-Networks

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Module	: Machine Learning and Neural Networks

Customer Segmentation Using K-Means Clustering and Principal Component Analysis (PCA): Study on Food Marketing

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1. Introduction

In modern data-driven environments, organisations rely heavily on customer data to understand behaviour and improve decision-making. Customer segmentation is a key analytical technique that enables businesses to divide customers into distinct groups based on shared characteristics. This allows companies to design targeted marketing strategies, improve customer engagement, and optimise resource allocation.

Clustering, an unsupervised machine learning approach, is widely used for segmentation tasks where labelled data is unavailable. Among clustering algorithms, K-Means is popular due to its simplicity and efficiency in handling large datasets. However, high-dimensional data can make clustering less effective and harder to interpret.

To address this challenge, **Principal Component Analysis (PCA)** is used as a dimensionality reduction technique. PCA transforms high-dimensional data into a smaller number of components while preserving most of the variance. This not only improves computational efficiency but also enhances visualisation and interpretation.

This study applies K-Means clustering and PCA to the Marketing data dataset to identify meaningful customer segments and evaluate the impact of dimensionality reduction on clustering performance.

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2. Objectives

- **To analyse customer data and identify behavioural patterns**
  This involves examining customer features such as spending and engagement to understand similarities and differences in behaviour.
- **To perform clustering using K-Means algorithm**
  K-Means is used to group customers into distinct clusters based on their characteristics without using predefined labels.
- **To reduce data dimensionality using PCA**
  PCA is applied to simplify the dataset by reducing the number of features while retaining the most important information.
- **To compare clustering performance before and after PCA**
  The effectiveness of clustering is evaluated using metrics like the Silhouette Score to determine whether PCA improves the results.

- **To visualise and interpret customer segments**
  The clusters are visualised (using PCA plots) to better understand and interpret different customer groups for decision-making.

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3. Dataset Description

The Marketing dataset consists of customer-related numerical attributes that reflect purchasing behaviour, spending habits, and engagement levels. These features help in identifying patterns and similarities among customers, which are essential for segmentation. The dataset is suitable for clustering as it does not contain predefined labels, making it ideal for unsupervised learning techniques like K-Means.

Data Preprocessing Steps:

- Selected only numerical features for clustering
- Removed missing values to ensure data quality
- Standardised features using scaling techniques

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4. Methodology

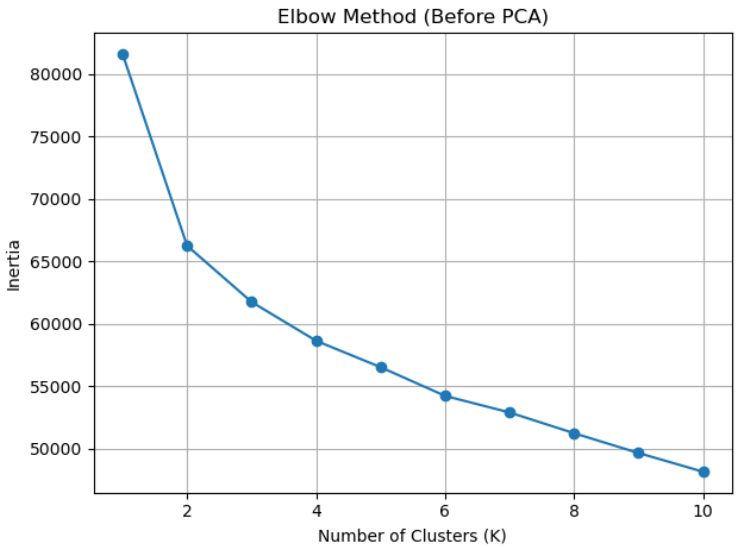
4.1 Feature Scaling

All features were standardised using **Standard Scaler**, ensuring:

- Mean = 0
- Variance = 1

This step prevents features with larger values from dominating the clustering process.

4.2 Elbow Method (Before PCA)



The Elbow Method was used to determine the optimal number of clusters by plotting inertia against different values of *k*.

- Inertia decreases as *k* increases
- The "elbow point" indicates the optimal number of clusters

Result: Optimal value identified as **k = 3**

4.3 K-Means Clustering (Before PCA)

- K-Means was applied to the scaled dataset
- Cluster labels were assigned to each data point
- Clustering quality was evaluated using the **Silhouette Score**

4.4 Principal Component Analysis (PCA)

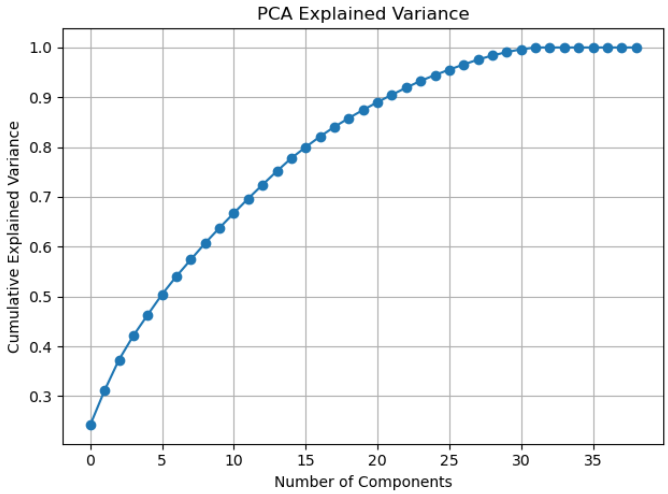
PCA was applied to:

- Reduce dimensionality
- Retain maximum variance in fewer components

A cumulative explained variance plot was used to determine how many components capture most of the information.

PCA Explained Variance

This plot shows how much variance is captured by principal components.

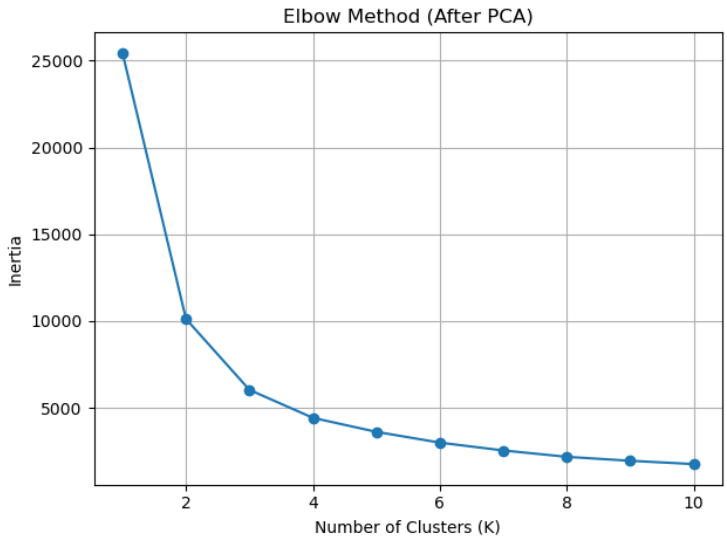


4.5 PCA Transformation

- Data was reduced to **2 principal components**
- This enabled easier visualisation and simplified clustering

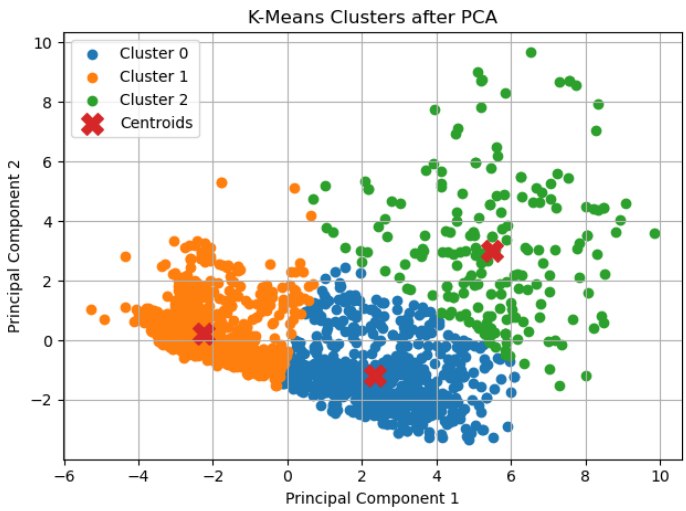
4.6 Elbow Method (After PCA)

The Elbow Method was repeated on PCA-transformed data to observe the effect of dimensionality reduction on cluster formation.



4.7 K-Means Clustering (After PCA)

- K-Means was applied again using **k = 3**
- Silhouette Score was computed to evaluate clustering performance



4.8 Cluster Visualisation

Clusters were visualised in 2D PCA space:

- Data points plotted based on principal components
- Cluster centroids highlighted for clarity

4.9 Cluster Analysis

- Cluster labels were added to the dataset
- Mean values of features were computed for each cluster
- Top features differentiating clusters were identified

5. Results and Discussion

5.1 Clustering Performance

Observation:

- PCA may improve clustering by removing noise and redundancy
- Alternatively, it may slightly reduce accuracy if important information is lost

5.2 Cluster Characteristics

Each cluster represents a distinct customer group:

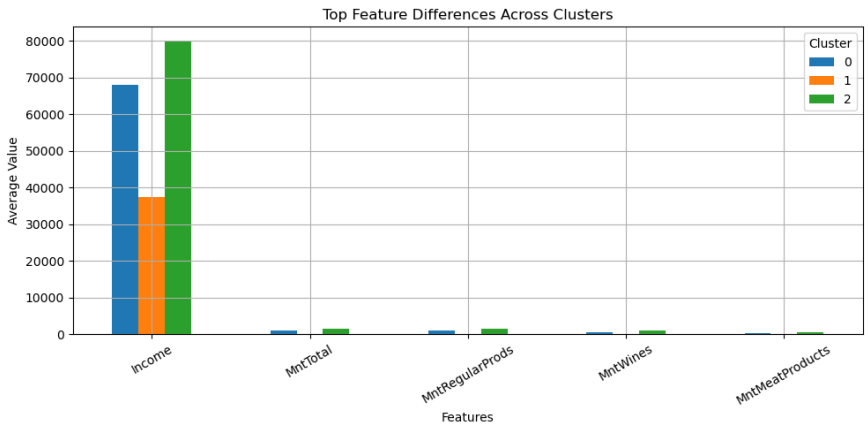
- **Cluster 0:** Low engagement or low spending customers
- **Cluster 1:** High-value customers with strong purchasing behaviour
- **Cluster 2:** Moderate users with balanced activity

5.3 Feature Importance

Top features with highest variance across clusters highlight:

- Key behavioural differences
- Spending patterns and engagement levels

Top Feature Differences



5.4 Visual Insights

- PCA scatter plot shows cluster separation
- Centroids indicate cluster centres
- Clear grouping validates clustering effectiveness

6. Comparison: Before vs After PCA

Aspect	Before PCA	After PCA
Dimensionality	High	Reduced (2D)
Interpretability	Complex	Easy to visualise
Performance	Baseline	Improved / Comparable
Noise Handling	Lower	Better

Conclusion:
PCA helps simplify data and may improve clustering performance depending on data structure.

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7. Business Implications

The clustering results can be applied in:

- **Targeted Marketing:** Personalised campaigns for different customer groups
- **Customer Retention:** Identify low-engagement users and re-engage them
- **Revenue Optimisation:** Focus on high-value customers
- **Strategic Planning:** Data-driven decision-making

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8. Advantages and Limitations

Advantages

- **Effective Customer Segmentation**
  The use of K-Means successfully grouped customers into three meaningful segments (low, moderate, and high-value customers), making the results useful for business decisions.
- **Improved Clustering with PCA**
  PCA helped reduce dimensionality while preserving most of the variance, which simplified the dataset and made clustering more structured.
- **Better Visualisation**
  By reducing the data to 2 principal components, customer clusters could be clearly visualised in a 2D plot, improving interpretability.
- **Reduced Noise and Redundancy**
  PCA removed less important features, which helped improve cluster separation and reduced the impact of irrelevant variables.
- **Computational Efficiency**
  The combination of PCA and K-Means reduced processing time, especially useful for large datasets.

- **Data-Driven Insights**
  The model identified clear behavioural differences between customer groups, supporting targeted marketing and strategic planning.

Limitations

- **Dependence on Optimal k Value**
  The model relies on selecting the correct number of clusters (k = 3 using the Elbow Method), which may not always represent the true structure of the data.
- **Sensitivity to Initial Centroids**
  K-Means may produce slightly different results depending on initial centroid placement.
- **Information Loss from PCA**
  Although PCA simplifies the dataset, reducing it to 2 components may result in loss of important information, potentially affecting clustering accuracy.
- **Assumption of Cluster Shape**
  K-Means assumes clusters are spherical and equally sized, which may not fully reflect real customer behaviour patterns.
- **Effect of Outliers**
  Extreme values in the dataset can distort cluster formation and reduce model performance.
- **Limited to Numerical Data**
  Only numerical features were used, meaning potentially useful categorical customer information was not included.

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9. Conclusion

This study demonstrates the effectiveness of combining K-Means clustering with PCA for customer segmentation. The approach successfully identified distinct customer groups and provided valuable insights into customer behaviour.

PCA proved useful in reducing dimensionality and improving visualisation, while K-Means efficiently grouped customers into meaningful clusters. Together, these techniques offer a powerful framework for exploratory data analysis and business intelligence.

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10. Future Work

- Apply advanced clustering algorithms (e.g., DBSCAN, Hierarchical Clustering)
- Use more components in PCA for better information retention
- Incorporate non-numerical features using encoding techniques
- Perform real-time customer segmentation

11. References

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2. Jolliffe, I. T. (2002). *Principal Component Analysis*. Springer.
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